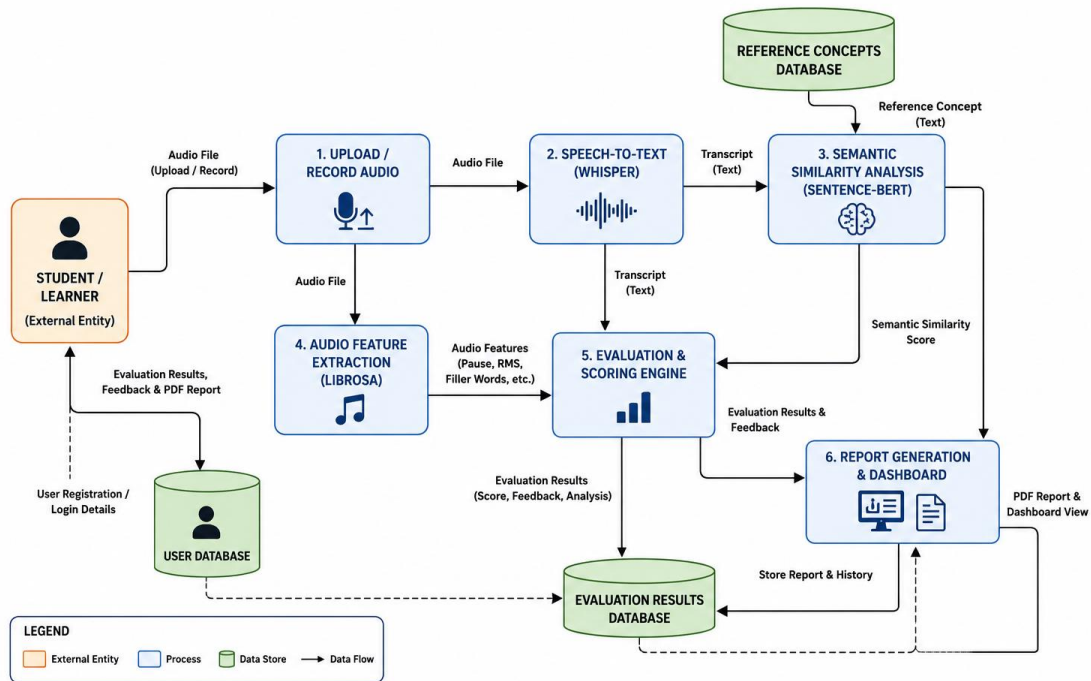


Data Flow Diagram

Date	27 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	2 Marks

Data Flow:

DATA FLOW DIAGRAM (DFD) – VOICE-BASED CONCEPT UNDERSTANDING ANALYSER (VBCUA)



Data Flow Description:

The **Voice-Based Concept Understanding Analyser (VBCUA)** Data Flow Diagram (DFD) illustrates how data moves through the system from audio input to intelligent

evaluation and report generation. The user uploads or records an audio explanation of a selected concept through the Streamlit interface. The audio is processed by the Speech-to-Text module using OpenAI Whisper to generate a transcript. The transcript is then compared with the reference concept using Sentence-BERT to calculate semantic similarity, while the audio is analyzed using Librosa to extract speech features such as pause ratio, RMS energy, and filler word usage. The evaluation engine combines these results to generate an overall understanding score and qualitative feedback. Finally, all evaluation results, transcripts, and reports are stored and displayed on the interactive dashboard, where users can also download a detailed PDF report.

DFD Symbol Legend:

Symbol	Name	Usage in Project
Oval	External Entity	Represents the Student/Learner, Teacher , or external AI models that provide input or receive evaluation results.
Rectangle	Process	Represents system processes such as User Authentication, Audio Upload, Speech-to-Text Transcription, Semantic Similarity Analysis, Audio Feature Extraction, Scoring Engine, and Report Generation.
Database/ Cylinder	Data Store	Stores User Information, Reference Concepts, Uploaded Audio Files, Transcripts, Evaluation Results, Audio Features, and Generated Reports.
Arrow	Data Flow	Shows the movement of data between the user, Streamlit application, Whisper module, Sentence-BERT engine, Audio Analysis module, databases, and the

		reporting system (e.g., audio files, transcripts, semantic scores, evaluation results, and PDF reports).
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